Lattice Boltzmann Formulation of the Magnetostatic Potential for a Non-Collinear (MRSh) Ferrofluid

$\nabla^2\psi = \nabla\cdot M$ with MRT collision and a single sub-step

Derivation, boundary closures, and error analysis

Abstract

The closure $\nabla\cdot(\mu\nabla\psi) = 0$ used in phase-field lattice Boltzmann ferrofluid models presupposes $\boldsymbol{M} \parallel \boldsymbol{H}$ and is inadmissible once Martsenyuk–Raikher–Shliomis magnetisation dynamics is retained, because $\boldsymbol{M} \times \boldsymbol{H} \neq 0$ is the entire content of the rotational-viscosity effect. This document derives a multiple-relaxation-time lattice Boltzmann scheme for the resulting Poisson problem $\nabla^2\psi = \nabla\cdot\boldsymbol{M}$, in which the magnetic field is obtained *entirely* from the lattice variable, $\boldsymbol{H} = \bar{\boldsymbol{H}} - \nabla\psi$. Section II establishes that the constant $\bar{\boldsymbol{H}}$ is not an implementation convenience but a topologically forced degree of freedom: on a domain periodic in the field direction, a single-valued potential carries zero mean field, so the applied field must be represented by $\bar{\boldsymbol{H}}$, exactly as a mean pressure gradient must be represented by a body force. On a domain bounded in the field direction $\bar{\boldsymbol{H}} \equiv 0$ and ψ is the total potential, the applied field entering solely through boundary data. The three free constants of the scheme (D, c_Φ, a) are determined rather than assumed; all three boundary closures are derived from discrete balances rather than calibrated; the discrete solvability condition is obtained as an exact identity; and the temporal response is characterised so that one sub-step per fluid step is provably sufficient. Error estimates are closed-form, with numerical values quoted only as sanity checks.

I. THE CONTINUUM PROBLEM

A. Magnetostatics

For a non-conducting magnetic fluid carrying no free currents, Maxwell's equations reduce to

$$\nabla\times\boldsymbol{H} = 0, \qquad \nabla\cdot\boldsymbol{B} = 0, \tag{1}$$

so that $\boldsymbol{H}$ is irrotational and $\boldsymbol{B}$ solenoidal. In SI units,

$$\boldsymbol{B} = \mu_0\,(\boldsymbol{H} + \boldsymbol{M})\,. \tag{2}$$

Irrotationality on a simply connected domain admits a single-valued scalar potential,

$$\boldsymbol{H} = -\nabla\psi, \tag{3}$$

and substituting (2) and (3) into $\nabla\cdot\boldsymbol{B} = 0$ gives

$$\boxed{\nabla^2\psi = \nabla\cdot\boldsymbol{M}} \tag{4}$$

a constant-coefficient Poisson equation. The permeability has disappeared, and with it every spatially varying coefficient; the source $\nabla\cdot\boldsymbol{M}$ is the density of bound magnetic charge.

B. Why the susceptibility closure is inadmissible

The reference phase-field model uses

$$\nabla\cdot(\mu\nabla\psi) = 0, \qquad \mu = \mu_0(1+\chi), \tag{5}$$

which is equivalent to (4) *if and only if* $\boldsymbol{M} = \chi \boldsymbol{H}$ with scalar χ, i.e. $\boldsymbol{M} \parallel \boldsymbol{H}$. The MRSh closure retained here,

$$\boldsymbol{M} = \phi\, \varphi_p M_s\, L(\xi_e)\, \frac{\boldsymbol{\xi}_e}{\xi_e}, \tag{6}$$

with $\boldsymbol{\xi}_e$ obeying its own relaxation dynamics, is non-collinear by construction: $\boldsymbol{M} \times \boldsymbol{H} \neq 0$ is precisely the torque producing $\Delta\eta$. Equation (5) therefore contradicts the magnetisation model and must be replaced by (4). Here ϕ is the phase-field order parameter and φ_p the particle volume fraction.

C. Source term in the two-phase case

Writing $\boldsymbol{M} = \phi\, \boldsymbol{M}_f$ with $\boldsymbol{M}_f$ the magnetisation of the pure ferrofluid phase,

$$\nabla \cdot \boldsymbol{M} = \phi\, \nabla \cdot \boldsymbol{M}_f + \boldsymbol{M}_f \cdot \nabla \phi. \tag{7}$$

The second term is the interfacial bound charge. It vanishes identically for $\phi \equiv 1$ but dominates once droplets are present, and is the reason the source must be evaluated with the same isotropic stencil already used for $\nabla\phi$.

II. REPRESENTATION OF THE APPLIED FIELD

Equation (3) asserts that *all* of $\boldsymbol{H}$ derives from ψ. Whether that is attainable depends on the topology of the domain, and the answer dictates the entire structure of the solver.

A. A topological obstruction

Theorem 1 (Mean-field obstruction)**.** *Let Ω be periodic in the direction $\hat{\boldsymbol{e}}$ with period L, and let $\boldsymbol{H} = -\nabla\psi$ with ψ single-valued and Ω-periodic. Then*

$$\left\langle \boldsymbol{H} \cdot \hat{\boldsymbol{e}} \right\rangle_\Omega = 0. \tag{8}$$

Proof. Let C be a straight line along $\hat{\boldsymbol{e}}$ closing on itself through the periodic identification. Then $\oint_C \boldsymbol{H} \cdot d\boldsymbol{\ell} = -\oint_C \nabla\psi \cdot d\boldsymbol{\ell} = -[\psi]_C = 0$ by single-valuedness and periodicity. Dividing by L gives the mean of $\boldsymbol{H} \cdot \hat{\boldsymbol{e}}$ along C; averaging over the transverse coordinates gives (8). □

A periodic scalar potential can therefore represent only fields of zero mean along every periodic direction. A uniform applied field along such a direction must be carried by a separate term.

Corollary 2 (Unique decomposition)**.** *Any curl-free $\boldsymbol{H}$ on such a domain admits the unique decomposition*

$$\boldsymbol{H} = \bar{\boldsymbol{H}}(t) - \nabla\psi, \qquad \bar{\boldsymbol{H}} = \left\langle \boldsymbol{H} \right\rangle_\Omega = \textit{const in } \boldsymbol{x}, \qquad \psi \textit{ periodic}, \quad \langle\psi\rangle = 0. \tag{9}$$

Proof. Existence: set $\bar{\boldsymbol{H}} = \langle \boldsymbol{H} \rangle$; then $\boldsymbol{H} - \bar{\boldsymbol{H}}$ is curl-free with zero mean and admits a periodic potential. Uniqueness: two such decompositions differ by a constant vector that is the mean of a periodic gradient, hence zero. □

Both pieces of (9) are determined: $\bar{\boldsymbol{H}}$ by the external source (a solenoid), ψ by (4) together with the transverse boundary conditions. Note that $\langle \nabla\psi \rangle = 0$ for periodic ψ, so $\bar{\boldsymbol{H}}$ is precisely the volume-averaged field and is measurable, not a fitting parameter.

B. Physical reading, and the exact hydrodynamic analogy

Equation (9) is the magnetostatic image of the pressure decomposition already used in the momentum solver. In a streamwise-periodic channel the pressure cannot carry a mean gradient, so one writes $p = -Gx + \tilde{p}$ with $\tilde{p}$ periodic and imposes G as a body force. Identically, the field cannot carry a mean along x through ψ, so one imposes it as $\bar{\boldsymbol{H}}$. Consequently *the additive $\bar{\boldsymbol{H}}$ term is not a modelling shortcut: it is the unique admissible representation*, and dropping it would force $\langle H_x \rangle = 0$, i.e. no applied field at all.

C. When the obstruction is absent

If Ω is bounded (non-periodic) in the field direction, no loop winds and the theorem does not apply. Then

$$\bar{\boldsymbol{H}} \equiv 0, \qquad \boldsymbol{H} = -\nabla\psi, \tag{10}$$

ψ is the total potential, and the applied field enters *entirely* through boundary data, e.g. $\psi = -\boldsymbol{H}_0 \cdot \boldsymbol{x}$ on the inflow/outflow faces. The lattice then carries the whole field. Both cases are covered below by retaining $\bar{\boldsymbol{H}}$ symbolically; (10) is the special case $\bar{\boldsymbol{H}} = 0$.

D. Conditioning

Even where (10) is admissible, the split is better conditioned. The applied part of the potential has magnitude $|\bar{\boldsymbol{H}}|L$ while the induced part scales as $|\boldsymbol{M}|\ell$, ℓ a structure length, giving a ratio

$$\frac{|\psi_{\text{applied}}|}{|\psi_{\text{induced}}|} \sim \frac{|\bar{\boldsymbol{H}}|L}{|\boldsymbol{M}|\ell}. \tag{11}$$

For the present channel this exceeds 10^5, so storing the total potential in one variable sacrifices five to six significant digits of the quantity actually wanted. Because $\psi_{\text{applied}} = -\bar{\boldsymbol{H}} \cdot \boldsymbol{x}$ is harmonic, removing it changes neither the source nor the interior equation: the split is exact, not approximate.

E. Boundary condition on ψ

At a fluid boundary with vacuum outside, the normal component of $\boldsymbol{B}$ is continuous. For a slab geometry the exterior induced field vanishes, so the exterior normal field is the applied one, and

$$\hat{\boldsymbol{n}} \cdot (\boldsymbol{H} + \boldsymbol{M})\Big|_{\text{in}} = \hat{\boldsymbol{n}} \cdot \boldsymbol{H}_0 \qquad \Longrightarrow \qquad \boxed{\frac{\partial \psi}{\partial n} = \gamma \equiv \hat{\boldsymbol{n}} \cdot \boldsymbol{M} - \hat{\boldsymbol{n}} \cdot \big(\boldsymbol{H}_0 - \bar{\boldsymbol{H}}\big)} \tag{12}$$

Two limits: with the split (9) and $\bar{\boldsymbol{H}} = \boldsymbol{H}_0$ this reduces to $\gamma = \hat{\boldsymbol{n}} \cdot \boldsymbol{M}$; with the total potential (10) it becomes $\gamma = \hat{\boldsymbol{n}} \cdot \boldsymbol{M} - \hat{\boldsymbol{n}} \cdot \boldsymbol{H}_0$. The condition is one of magnetic insulation of the *induced* field, not of the total field.

III. PSEUDO-TIME EMBEDDING AND LATTICE PRELIMINARIES

A. Embedding

A lattice Boltzmann equation cannot integrate an elliptic operator directly, so (4) is embedded as the steady state of a diffusion equation in a fictitious time t:

$$\frac{\partial \psi}{\partial t} = D\nabla^2\psi + S, \qquad S \equiv -D\nabla \cdot \boldsymbol{M}. \tag{13}$$

Setting $\partial_t \psi = 0$ returns (4) for *any* $D > 0$, because the same D multiplies both terms. This is the load-bearing property: if the D multiplied into S differs from the D the lattice realises, the steady state becomes $\nabla^2\psi = (D_{\text{code}}/D_{\text{lattice}})\nabla\cdot\boldsymbol{M}$, a silent multiplicative error on every field. Determining D_{lattice} exactly for the MRT operator is therefore the first task.

B. Velocity set and moment identities

The D2Q9 set is used, with $c = \Delta x/\Delta t$ and $c_s^2 = c^2/3$:

$$\boldsymbol{e}_k = \begin{cases} (0,0), & k = 0, \\ c\left(\cos\frac{(k-1)\pi}{2},\ \sin\frac{(k-1)\pi}{2}\right), & k = 1,\dots,4, \\ \sqrt{2}\,c\left(\cos\frac{(2k-1)\pi}{4},\ \sin\frac{(2k-1)\pi}{4}\right), & k = 5,\dots,8, \end{cases} \qquad w_k = \tfrac{4}{9},\ \tfrac{1}{9},\ \tfrac{1}{36}. \tag{14}$$

Four weight identities follow from antipodal symmetry and are all that is used below:

$$\sum_k w_k = 1, \quad \sum_k w_k e_{k\alpha} = 0, \quad \sum_k w_k e_{k\alpha} e_{k\beta} = c_s^2 \delta_{\alpha\beta}, \quad \sum_k w_k e_{k\alpha} e_{k\beta} e_{k\gamma} = 0. \tag{15}$$

C. Equilibrium

The potential is not advected, so the equilibrium carries no velocity coupling:

$$h_k^{\mathrm{eq}} = w_k\,\psi, \tag{16}$$

with moments, from (15),

$$\sum_k h_k^{\mathrm{eq}} = \psi, \quad \sum_k e_{k\alpha} h_k^{\mathrm{eq}} = 0, \quad \sum_k e_{k\alpha} e_{k\beta} h_k^{\mathrm{eq}} = c_s^2\,\psi\,\delta_{\alpha\beta}, \quad \sum_k e_{k\alpha} e_{k\beta} e_{k\gamma} h_k^{\mathrm{eq}} = 0. \tag{17}$$

The vanishing first moment removes advection; the second moment becomes the Laplacian; the vanishing third moment closes the expansion at second order.

D. Source and macroscopic definition, with undetermined constants

$$\Phi_k = c_\Phi\, w_k\, S, \qquad \psi \equiv \sum_k h_k + a\,S, \qquad \sum_k \Phi_k = c_\Phi S, \quad \sum_k e_{k\alpha}\Phi_k = 0. \tag{18}$$

No assumption $c_\Phi = 1$, $a = 0$ is made; both are derived in Sec. 5.

E. MRT evolution

With M the orthogonal moment matrix of the reference model,

$$\Lambda = \mathsf{M}^{-1}\mathsf{S}_h\mathsf{M}, \qquad \mathsf{S}_h = \mathrm{diag}\left(s_0,\, s_1',\, s_2',\, s_1,\, s_1,\, s_1,\, s_1,\, s_7',\, s_8'\right), \tag{19}$$

where $s_1 \equiv s_3 = s_5$ is the first-moment rate, the evolution equation is

$$h_k(\boldsymbol{x} + \boldsymbol{e}_k\Delta t, t + \Delta t) - h_k(\boldsymbol{x}, t) = -\Lambda_{k\ell}\left[h_\ell - h_\ell^{\mathrm{eq}}\right](\boldsymbol{x}, t) + \Delta t\,\Phi_k(\boldsymbol{x}, t). \tag{20}$$

IV. THE MRT PROJECTION LEMMA

The analysis of (20) requires exactly two contractions of Λ. Both follow from the structure of M: row 0 is $M_{0k} = 1$, row 3 is $M_{3k} = e_{kx}/c$, row 5 is $M_{5k} = e_{ky}/c$. Using $\sum_k M_{ak}(\mathsf{M}^{-1})_{kb} = \delta_{ab}$,

$$\sum_k \Lambda_{k\ell} = \sum_a s_a M_{a\ell} \sum_k (\mathsf{M}^{-1})_{ka} = \sum_a s_a M_{a\ell}\,\delta_{0a} = s_0, \tag{21}$$

$$\sum_k e_{k\alpha}\Lambda_{k\ell} = c\sum_a s_a M_{a\ell} \sum_k M_{(\alpha)k}(\mathsf{M}^{-1})_{ka} = s_1\, e_{\ell\alpha}, \qquad (\alpha) = 3, 5. \tag{22}$$

In words: *the zeroth moment relaxes at s_0 and the first moment at s_1; no other relaxation rate enters the recovered macroscopic equation.* The rates s_1', s_2', s_7', s_8' act on ghost modes only and remain free. The lemma reduces the MRT algebra to the BGK case, to which it collapses when $s_0 = s_1 = 1/\tau$.

V. CHAPMAN–ENSKOG ANALYSIS

A. Taylor expansion

With the link derivative $\mathscr{D}_k \equiv \partial_t + \boldsymbol{e}_k \cdot \nabla$, expanding the streaming operator and dividing by Δt,

$$\mathscr{D}_k h_k + \frac{\Delta t}{2}\mathscr{D}_k^2 h_k + O(\Delta t^2) = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{\mathrm{neq}} + \Phi_k, \qquad h_k^{\mathrm{neq}} \equiv h_k - h_k^{\mathrm{eq}}. \tag{23}$$

The term $\frac{\Delta t}{2}\mathscr{D}_k^2 h_k$ is retained: it is the origin of the $-\Delta t/2$ in the diffusivity.

B. Expansion and ordering

$$h_k = h_k^{(0)} + \delta h_k^{(1)} + \delta^2 h_k^{(2)}, \quad \partial_t = \delta\partial_{t_1} + \delta^2\partial_{t_2}, \quad \nabla = \delta\nabla_1, \quad \Phi_k = \delta\Phi_k^{(1)}, \quad S = \delta S_1. \tag{24}$$

Ordering the source at $O(\delta)$ is the general case and is what determines a and c_Φ uniquely. Collecting orders in (23):

$$O(\delta^0): \quad h_k^{(0)} = h_k^{\mathrm{eq}} = w_k\psi, \tag{25}$$

$$O(\delta^1): \quad \Delta t\,\mathscr{D}_k^{(1)} h_k^{(0)} = -\Lambda_{k\ell} h_\ell^{(1)} + \Delta t\,\Phi_k^{(1)}, \tag{26}$$

$$O(\delta^2): \quad \partial_{t_2} h_k^{(0)} + \mathscr{D}_k^{(1)} h_k^{(1)} + \frac{\Delta t}{2}\left(\mathscr{D}_k^{(1)}\right)^2 h_k^{(0)} = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{(2)}. \tag{27}$$

Applying $\mathscr{D}_k^{(1)}$ to (26) and substituting removes the second derivative:

$$\partial_{t_2} h_k^{(0)} + \mathscr{D}_k^{(1)} h_k^{(1)} - \frac{1}{2}\mathscr{D}_k^{(1)}\left[\Lambda_{k\ell} h_\ell^{(1)}\right] + \frac{\Delta t}{2}\mathscr{D}_k^{(1)}\Phi_k^{(1)} = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{(2)}. \tag{28}$$

Expanding $\psi = \sum_k h_k + aS$ order by order gives the solvability constraints

$$\sum_k h_k^{(1)} = -a S_1, \qquad \sum_k h_k^{(2)} = 0. \tag{29}$$

C. Moment balances

First moment of (26), using (17), (18) and lemma (22), with $J_\alpha \equiv \sum_k e_{k\alpha} h_k^{(1)}$:

$$\Delta t\, c_s^2\,\partial_{1\alpha}\psi = -s_1 J_\alpha \qquad \Longrightarrow \qquad J_\alpha = -\frac{\Delta t\, c_s^2}{s_1}\,\partial_{1\alpha}\psi. \tag{30}$$

Zeroth moment of (26), using (29) and lemma (21):

$$\partial_{t_1}\psi = \left(\frac{s_0 a}{\Delta t} + c_\Phi\right) S_1. \tag{31}$$

Zeroth moment of (28), using both lemmas, (29) and (30):

$$\partial_{t_2}\psi = \Delta t\, c_s^2\left(\frac{1}{s_1} - \frac{1}{2}\right)\nabla_1^2\psi + \left[a\left(1 - \frac{s_0}{2}\right) - \frac{\Delta t}{2}c_\Phi\right]\partial_{t_1} S_1. \tag{32}$$

D. Recovered equation and closure of the constants

Recombining $\partial_t = \delta\partial_{t_1} + \delta^2\partial_{t_2}$,

$$\boxed{\partial_t\psi = \underbrace{\Delta t\, c_s^2\left(\frac{1}{s_1} - \frac{1}{2}\right)}_{D}\nabla^2\psi + \underbrace{\left(\frac{s_0 a}{\Delta t} + c_\Phi\right)}_{\mathscr{C}} S + \underbrace{\left[a\left(1 - \frac{s_0}{2}\right) - \frac{\Delta t}{2}c_\Phi\right]}_{\mathscr{T}}\partial_t S} \tag{33}$$

Matching to (13) gives

$$\text{(C1) diffusivity:} \quad D = \Delta t\, c_s^2\left(\frac{1}{s_1} - \frac{1}{2}\right) = \Delta t\, c_s^2\left(\tau_h - \frac{1}{2}\right), \quad \tau_h \equiv 1/s_1, \tag{34}$$

$$\text{(C2) source normalisation:} \quad \mathscr{C} = \frac{s_0 a}{\Delta t} + c_\Phi = 1, \tag{35}$$

$$\text{(C3) transient removal:} \quad \mathscr{T} = a\left(1 - \frac{s_0}{2}\right) - \frac{\Delta t}{2}c_\Phi = 0. \tag{36}$$

Three consequences.

(i) D depends on s_1 *only*; the zeroth-moment rate s_0 does not enter it. The BGK formula $D = c_s^2(\tau - \frac{1}{2})\Delta t$ remains valid in MRT provided τ is read as $1/s_3 = 1/s_5$.

(ii) (C2) and (C3) together have a unique solution. Substituting $c_\Phi = 1 - s_0 a/\Delta t$ into (C3), the terms $-s_0 a/2$ cancel and

$$a = \frac{\Delta t}{2}, \qquad c_\Phi = 1 - \frac{s_0}{2}. \tag{37}$$

(iii) Only (C2) fixes the steady state, since $\partial_t S \to 0$ there. A second admissible closure therefore exists,

$$a = 0, \qquad c_\Phi = 1 \qquad \Longrightarrow \qquad \mathscr{C} = 1, \quad \mathscr{T} = -\frac{\Delta t}{2}, \tag{38}$$

exact at steady state but retaining a residual $-\frac{\Delta t}{2}\partial_t S$. Section 10 shows (38) is the more accurate of the two in the quasi-static regime enforced here. Any other pair violates (C2) and rescales every field by $\mathscr{C} \neq 1$.

VI. RECOVERY OF THE MAGNETIC FIELD

Since $\sum_k \boldsymbol{e}_k h_k^{\mathrm{eq}} = 0$, the first moment of h is purely non-equilibrium, and (30) inverts. Combining with the decomposition (9),

$$\boxed{H_\alpha = \bar{H}_\alpha(t) + \frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{k\alpha} h_k} \tag{39}$$

The second term is $-\partial_\alpha \psi$ in its entirety: the induced field is obtained from the lattice alone, with no auxiliary gradient stencil and no ghost nodes at the walls. The first term is the topologically forced mean field of Sec. 2; it vanishes identically, $\bar{\boldsymbol{H}} = 0$, whenever the domain is bounded in the field direction, in which case (39) delivers the total field from the lattice.

The leading correction to (39) follows from the first moment of (28):

$$\sum_k e_{k\alpha} h_k^{(2)} = \tau_h^2 c_s^2 \left(1 - \frac{s_1}{2}\right) \partial_{1\alpha}(\partial_{t_1}\psi) \;\propto\; \partial_\alpha S, \tag{40}$$

which vanishes identically at convergence, $\partial_t \psi \to 0$. Equation (39) is therefore exact in the quasi-static limit quantified in Sec. 9.

VII. BOUNDARY CLOSURES FOR h

Three closures are required. Each is derived from a discrete balance, not calibrated. Throughout, $\boldsymbol{x}_b$ is the boundary node, the wall lies halfway to its image, $\hat{\boldsymbol{n}}$ is the outward unit normal, and

$$K_{\mathrm{in}} = \{k : \; \boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} < 0\}, \qquad W_{\mathrm{in}} = \sum_{k \in K_{\mathrm{in}}} w_k, \qquad \boldsymbol{e}_{\sigma(k)} = \boldsymbol{e}_k - 2(\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}}, \qquad \boldsymbol{e}_{\bar{k}} = -\boldsymbol{e}_k, \tag{41}$$

σ being specular reflection and $\bar{k}$ the opposite direction. Note $w_{\sigma(k)} = w_{\bar{k}} = w_k$.

A. Neumann closure (magnetic insulation)

Pure specular reflection, $h_k(\boldsymbol{x}_b, t+\Delta t) = h^*_{\sigma(k)}(\boldsymbol{x}_b, t)$ for $k \in K_{\mathrm{in}}$, transfers

$$\sum_{k \in K_{\mathrm{in}}} h_k(t+\Delta t) - \sum_{k \in K_{\mathrm{out}}} h_k^*(t) = 0, \tag{42}$$

because σ is a weight-preserving bijection $K_{\mathrm{in}} \leftrightarrow K_{\mathrm{out}}$. Specular reflection therefore imposes exactly $\partial\psi/\partial n = 0$, *not* magnetic insulation. Augmenting it,

$$h_k(\boldsymbol{x}_b, t+\Delta t) = h^*_{\sigma(k)}(\boldsymbol{x}_b, t) + B\, w_k, \qquad k \in K_{\mathrm{in}}, \tag{43}$$

transfers $BW_{\rm in}$ per step. The macroscopic flux is Fickian, $\boldsymbol{J} = -D\nabla\psi$, so the potential entering unit wall area per unit time is $-\hat{\boldsymbol{n}}\cdot\boldsymbol{J} = D\,\partial\psi/\partial n = D\gamma$. Equating,

$$\boxed{B = \frac{\Delta t\, D\,\gamma}{W_{\rm in}}, \qquad \gamma = \hat{\boldsymbol{n}}\cdot\boldsymbol{M} - \hat{\boldsymbol{n}}\cdot\left(\boldsymbol{H}_0 - \bar{\boldsymbol{H}}\right)} \tag{44}$$

For a flat D2Q9 wall three links enter, $W_{\rm in} = \frac{1}{9}+\frac{1}{36}+\frac{1}{36} = \frac{1}{6}$, so $B = 6\,\Delta t\, D\,\gamma$. For the channel, with $\hat{\boldsymbol{n}} = \mp\hat{\boldsymbol{y}}$,

$$\gamma_i^{\rm bot} = -M_y^{w,\rm bot}, \qquad \gamma_i^{\rm top} = +M_y^{w,\rm top}, \qquad M_y^{w} = \tfrac{3}{2}M_y\big|_{\text{1st node}} - \tfrac{1}{2}M_y\big|_{\text{2nd node}}, \tag{45}$$

the extrapolation placing γ at the wall rather than the node, which is required for second order. Setting $\gamma = 0$ recovers pure specular reflection, confirming consistency.

B. Dirichlet closure (prescribed potential)

When the field direction is bounded, the applied field is imposed as $\psi_w = -\boldsymbol{H}_0\cdot\boldsymbol{x}_w$ on the corresponding faces. The closure is anti-bounce-back,

$$\boxed{h_k(\boldsymbol{x}_b, t+\Delta t) = -h_{\bar{k}}^{*}(\boldsymbol{x}_b, t) + 2\,w_k\,\psi_w} \qquad k \in K_{\rm in}, \tag{46}$$

whose consistency follows from the equilibrium structure: for $h_k^{\rm eq} = w_k\psi$, the combination $h_k + h_{\bar{k}}$ carries the even (potential-bearing) part and $h_k - h_{\bar{k}}$ the odd (flux-bearing) part, so reversing the sign of the reflected population fixes the even part at $2w_k\psi_w$ while leaving the flux free. This is the standard advection–diffusion anti-bounce-back specialised to a source-free equilibrium; it is second-order accurate at a halfway wall.

C. Periodic and quasi-periodic closure

On a periodic face the populations stream unchanged. If instead the total potential is carried on a domain periodic in the field direction, single-valuedness fails and the closure becomes quasi-periodic,

$$h_k(\boldsymbol{x}_b, t+\Delta t) = h_k^{*}(\boldsymbol{x}_{\rm image}, t) + w_k\,\Delta\psi_{\rm per}, \qquad \Delta\psi_{\rm per} = -\bar{\boldsymbol{H}}\cdot\boldsymbol{L}, \tag{47}$$

$\boldsymbol{L}$ the period vector. Equation (47) is the lattice statement of the obstruction of Sec. 2: the jump $\Delta\psi_{\rm per}$ is exactly the circulation that a single-valued potential cannot carry. It is implementable, but since $|\Delta\psi_{\rm per}| = |\bar{\boldsymbol{H}}|L$ is larger than the induced potential by the ratio (11), the decomposition (9) with plain periodicity is preferred on conditioning grounds.

D. Interior and curved boundaries

Equations (44) and (46) are stated for general $\hat{\boldsymbol{n}}$ and carry over unchanged to obstacle surfaces; only $W_{\rm in}$ and $\hat{\boldsymbol{n}}$ change, and $W_{\rm in}$ must be evaluated per node from the actual incoming set rather than assumed equal to $1/6$ (corners differ). Curved boundaries additionally require γ or ψ_w at the true wall intersection, i.e. an interpolation weight replacing the fixed $\frac{3}{2}, -\frac{1}{2}$ of (45).

VIII. DISCRETE SOLVABILITY

Let $\Psi = \sum_{\boldsymbol{x}}\sum_k h_k$. Summing the collision over k and using lemma (21) with $\sum_k h_k - \psi = -aS$,

$$\sum_k h_k^{*} = \sum_k h_k + \left(s_0 a + \Delta t\, c_\Phi\right)S = \sum_k h_k + \Delta t\,\mathscr{C}\,S = \sum_k h_k + \Delta t\,S, \tag{48}$$

the last step by (C2). Streaming conserves Ψ except at Neumann faces, which inject $\Delta t\,D\gamma$ per node by (43)–(44). Hence, *if every face is Neumann or periodic*, a steady state exists if and only if

$$\boxed{\sum_{\boldsymbol{x}}\left(\nabla\cdot\boldsymbol{M}\right) = \sum_{\rm faces}\gamma} \tag{49}$$

the discrete image of the divergence theorem. Note that D cancels and c_Φ does not appear: the condition is a property of the discretisation alone. The solution is then unique only up to an additive constant, immaterial because only $\nabla\psi$ enters (39).

If any face carries the Dirichlet closure (46), the operator is non-singular, the constant is fixed, and (49) is not required. This is a structural advantage of the bounded-domain formulation.

In the all-Neumann case the isotropic nine-point stencil used for $\nabla\cdot\boldsymbol{M}$ and the extrapolation (45) are different approximations, so (49) holds only to truncation order. The residual is removed by projecting the source onto the range of the discrete Laplacian:

$$\Delta \equiv \sum_{\boldsymbol{x}} (\nabla\cdot M) - \sum_{\text{faces}} \gamma, \qquad (\nabla\cdot M) \longleftarrow (\nabla\cdot M) - \frac{\Delta}{N}, \tag{50}$$

N the number of fluid nodes. This is the standard treatment of the singular pure-Neumann pressure-Poisson system, where projection is preferred to pinning a Dirichlet value at one node. The ratio Δ/N must be monitored: it is a truncation diagnostic when γ is set consistently with (44), but becomes an $O(1)$ fictitious magnetic charge if γ is set to zero in violation of (49).

IX. TEMPORAL RESPONSE AND THE SINGLE-SUB-STEP CONDITION

Linearising (13) for a Fourier mode of wavenumber k driven at frequency ω,

$$\hat{\psi} = \frac{\hat{S}}{\mathrm{i}\omega + Dk^2} = \frac{\hat{\psi}_{\text{qs}}}{1+\mathrm{i}\,\omega\,\tau_{\text{relax}}}, \qquad \hat{\psi}_{\text{qs}} = \frac{\hat{S}}{Dk^2}, \tag{51}$$

$\hat{\psi}_{\text{qs}}$ being the true magnetostatic response. The lattice relaxation time is defined spectrally,

$$\tau_{\text{relax}} \equiv -\frac{\Delta t}{\ln|\lambda_2|}, \tag{52}$$

λ_2 the second-largest eigenvalue modulus of the one-step operator of (20), and possesses two asymptotic branches,

$$\tau_{\text{relax}} \sim \frac{1}{D k_{\text{min}}^2} \ \text{(hydrodynamic)}, \qquad \tau_{\text{relax}} \sim \tau_h = \frac{\varepsilon}{c_s^2} + \frac{1}{2} \ \text{(kinetic)}, \qquad D = \varepsilon. \tag{53}$$

The branches cross near $\varepsilon \simeq c_s/k_{\text{min}}$, so τ_{relax} has a *minimum* in ε: raising the relaxation parameter past that point makes the solver slower, not faster, and there is a floor no choice of ε can defeat. A single sub-step per fluid step is sufficient if and only if

$$\boxed{\omega\,\tau_{\text{relax}} \ll 1 \qquad \Longleftrightarrow \qquad \tau_B \gg (\omega\tau_B)\,\tau_{\text{relax}}} \tag{54}$$

since $\omega = (\omega\tau_B)/\tau_B$ and τ_B in lattice steps is a free nondimensionalisation choice. Condition (54) is not a numerical convenience: it states that the numerical magnetostatic relaxation must be fast compared with every physical time scale, which is the faithful lattice representation of instantaneous magnetostatics.

X. ERROR ANALYSIS

Four error sources are separated. Each is given in closed form; the numerical column of Table 1 is a sanity check only, obtained from exact steady-state linear solves and eigenvalue computations of the scheme itself.

A. E_1: spatial truncation

Truncation of the Chapman–Enskog hierarchy at $O(\delta^3)$ leaves a fourth-derivative remainder, so the steady solution is second-order accurate,

$$E_1 = \frac{\|\psi_h - \psi\|_\infty}{\|\psi\|_\infty} = C_1\,(k\Delta x)^2 + O\big((k\Delta x)^4\big). \tag{55}$$

The constant depends on the closure: since the shift contributes $a|S| = \frac{\Delta t}{2} Dk^2 |\psi|$ directly to the reported potential,

$$C_1\big(a = \tfrac{\Delta t}{2}\big) - C_1\big(a = 0\big) = \frac{D\,\Delta t}{2}. \tag{56}$$

B. E_2: temporal lag

From (51), exactly,

$$E_2^{\text{amp}} = 1 - \frac{1}{\sqrt{1 + (\omega\tau_{\text{relax}})^2}} \simeq \frac{(\omega\tau_{\text{relax}})^2}{2}, \qquad E_2^{\text{phase}} = \arctan\big(\omega\,\tau_{\text{relax}}\big). \tag{57}$$

Closure (38) carries an additional transient error of relative size $|\mathscr{T}|\,\omega = \omega\Delta t/2$.

C. E_3: source normalisation

Violating (C2) rescales the steady state exactly:

$$\nabla^2\psi = \mathscr{C}\,\nabla\cdot\boldsymbol{M}, \qquad E_3 = |\mathscr{C} - 1| = \left|\frac{s_0 a}{\Delta t} + c_\Phi - 1\right|. \tag{58}$$

D. E_4: compatibility defect

The projection (50) shifts the source uniformly, i.e. adds a fictitious uniform magnetic charge of density Δ/N:

$$E_4 = \frac{|\Delta|/N}{|\nabla\cdot\boldsymbol{M}|}. \tag{59}$$

E_4 is identically zero when a Dirichlet face is present, since (49) is then not imposed.

E. Selection between the two admissible closures

Combining (56) with the transient term of (38),

$$E\big(a = \tfrac{\Delta t}{2}\big) - E\big(a = 0\big) \;\simeq\; \frac{\Delta t}{2}\Big(Dk^2 - \omega\Big), \tag{60}$$

so that

$$\boxed{a = 0,\ c_\Phi = 1 \ \text{ is preferable whenever } \ \omega < Dk^2} \tag{61}$$

which is precisely the quasi-static condition (54) already imposed. The half-source shift is needed only when the drive approaches the lattice relaxation rate; in the regime enforced here it adds more spatial error than the transient error it removes.

F. Numerical sanity check

Two rows deserve comment. First, $\gamma \equiv 0$: because $\nabla\cdot\boldsymbol{M} = \partial_y M_y$ is single-signed across a channel, imposing $\partial\psi/\partial n = 0$ forces the projection (50) to remove the entire mean of the source, $E_4 = 1$. Second, the spectral radius of the one-step operator was found to be exactly unity in every case tested — the conserved mode — so the floor in τ_{relax} is an accuracy limit, never a stability limit.

XI. SUMMARY OF GOVERNING EQUATIONS

$$\boldsymbol{H} = \bar{\boldsymbol{H}}(t) - \nabla\psi, \qquad \text{field, mean part plus induced part} \tag{62}$$

Quantity	Prediction	Measured	Test
D vs. (34)	$\Delta t\, c_s^2(1/s_1 - \frac{1}{2})$, indep. of s_0	ratio 0.998–1.0001, $s_1 \in [0.08, 1.5]$	D-free source
Order of E_1	2	2.00 throughout $n_Y = 16 \to 128$	manufactured ψ
C_1 gap, (56)	$D\Delta t/2 = 2.5$ at $\varepsilon = 5$	$3.771 - 1.045 = 2.73$	manufactured ψ
E_3, closures (37), (38)	0	≤ 0.036 (i.e. $= E_1$)	manufactured ψ
E_3, $(c_\Phi = 1 - \frac{s_0}{2}, a = 0)$	0.500	0.495	manufactured ψ
E_3, $(c_\Phi = 1, a = \frac{\Delta t}{2})$	0.500	0.541	manufactured ψ
Neumann closure (44)	exact	rel. err 2.6×10^{-15}	exact quadratic
Dirichlet closure (46)	2nd order	2.00, $n_Y = 16 \to 128$	exact quadratic
Dirichlet, wrong coefficient ($w_k \psi_w$)	inconsistent	rel. err 0.31	exact quadratic
Solvability, γ from (44)	$E_4 \to 0$	3×10^{-22}	slab $M_y(y)$
Solvability, $\gamma \equiv 0$ imposed	$E_4 = O(1)$	$E_4 = 1.00$	slab $M_y(y)$
τ_{relax} minimum, (53)	at $\varepsilon \simeq c_s/k_{\min}$	min at $\varepsilon = 5$, $\tau_{\text{relax}} = 30$	spectral radius

Table 1: Closed-form predictions against numerical measurement. All measurements come from exact linear solves of the scheme, not from the production code.

$$\bar{\boldsymbol{H}} = \langle \boldsymbol{H} \rangle_\Omega \;\; (= \boldsymbol{H}_0 \text{ if periodic; } = 0 \text{ if bounded}), \qquad \text{topologically forced datum} \tag{63}$$

$$\nabla^2 \psi = \nabla \cdot \boldsymbol{M}, \qquad \text{magnetostatic potential} \tag{64}$$

$$\nabla \cdot \boldsymbol{M} = \phi\, \nabla \cdot \boldsymbol{M}_f + \boldsymbol{M}_f \cdot \nabla \phi, \qquad \text{bulk plus interfacial charge} \tag{65}$$

$$\frac{\partial \psi}{\partial t} = D \nabla^2 \psi + S, \qquad S = -D \nabla \cdot \boldsymbol{M}, \qquad \text{pseudo-time embedding} \tag{66}$$

$$h_k^{\text{eq}} = w_k \psi, \qquad \text{equilibrium} \tag{67}$$

$$\Phi_k = c_\Phi w_k S, \qquad \psi = \sum_k h_k + aS, \qquad \text{source and macroscopic potential} \tag{68}$$

$$h_k(\boldsymbol{x} + \boldsymbol{e}_k \Delta t, t + \Delta t) = h_k - \left(\mathsf{M}^{-1} \mathsf{S}_h \mathsf{M}\right)_{k\ell} \left(h_\ell - h_\ell^{\text{eq}}\right) + \Delta t\, \Phi_k, \qquad \text{MRT evolution} \tag{69}$$

$$D = \Delta t\, c_s^2 (1/s_1 - 1/2), \quad s_1 = s_3 = s_5, \qquad \text{lattice diffusivity} \tag{70}$$

$$s_0 a/\Delta t + c_\Phi = 1, \qquad \text{source normalisation (mandatory)} \tag{71}$$

$$a\,(1 - s_0/2) = \tfrac{\Delta t}{2} c_\Phi, \qquad \text{transient removal (optional)} \tag{72}$$

$$H_\alpha = \bar{H}_\alpha(t) + \frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{k\alpha} h_k, \qquad \text{field recovery from } h \tag{73}$$

$$h_k = h^*_{\sigma(k)} + B\, w_k, \quad B = \Delta t D \gamma / W_{\text{in}}, \qquad \text{Neumann closure} \tag{74}$$

$$\gamma = \hat{\boldsymbol{n}} \cdot \boldsymbol{M} - \hat{\boldsymbol{n}} \cdot (\boldsymbol{H}_0 - \bar{\boldsymbol{H}}), \qquad \text{insulation of the induced field} \tag{75}$$

$$h_k = -h^*_{\bar{k}} + 2 w_k \psi_w, \qquad \text{Dirichlet closure} \tag{76}$$

$$h_k = h^*_k(\boldsymbol{x}_{\text{image}}) + w_k \Delta \psi_{\text{per}}, \qquad \text{(quasi-)periodic closure} \tag{77}$$

$$\sum_{\boldsymbol{x}} (\nabla \cdot \boldsymbol{M}) = \sum_{\text{faces}} \gamma, \qquad \text{solvability (all-Neumann only)} \tag{78}$$

$$\omega\, \tau_{\text{relax}} \ll 1, \qquad \text{validity of one sub-step} \tag{79}$$

XII. IMPLEMENTABLE FORM

Lattice units: $\Delta x = \Delta t = 1$, $c = 1$, $c_s^2 = 1/3$, $\mu_0 = 1$. Indices (i, j), i periodic over n_X, $j = 0, \ldots, n_Y - 1$ between halfway walls.

A. Relaxation set

$$s_1 = \frac{1}{\varepsilon/c_s^2 + 1/2}, \qquad \tau_h = \frac{1}{s_1} = \frac{\varepsilon}{c_s^2} + \frac{1}{2}, \qquad D = c_s^2\left(\tau_h - \tfrac{1}{2}\right) = \varepsilon, \tag{80}$$

$$\mathsf{S}_h = \mathrm{diag}\big(s_0,\, 1.1,\, 1.1,\, s_1,\, s_1,\, s_1,\, s_1,\, 1.2,\, 1.2\big), \qquad s_0 = 1.0. \tag{81}$$

No permeability appears: ε is a pseudo-time relaxation rate of dimension $[\mathrm{L}^2\mathrm{T}^{-1}]$, and a factor μ_0 carried over from the susceptibility closure is dimensionally inconsistent with (4).

B. Closure constants

Adopting (38), justified by (61),

$$c_\Phi = 1, \qquad a = 0 \qquad \Longrightarrow \qquad \psi_{i,j} = \sum_k h_{k,i,j}, \qquad \Phi_k = w_k S_{i,j}. \tag{82}$$

C. Mean field and wall data

$$\bar{H}_x(t) = H_{0x}\cos(\omega t), \qquad \bar{H}_y(t) = H_{0y}\cos(\omega t) \qquad (x \text{ periodic} \Rightarrow \bar{\boldsymbol{H}} = \boldsymbol{H}_0), \tag{83}$$

$$\gamma_i^{\mathrm{bot}} = -\left(\tfrac{3}{2}M_{y,i,0} - \tfrac{1}{2}M_{y,i,1}\right), \qquad \gamma_i^{\mathrm{top}} = +\left(\tfrac{3}{2}M_{y,i,n_Y-1} - \tfrac{1}{2}M_{y,i,n_Y-2}\right). \tag{84}$$

If x is later made non-periodic, set $\bar{\boldsymbol{H}} = 0$, impose (46) on the x-faces with $\psi_w = -H_{0x}x_w$, and add $-\hat{\boldsymbol{n}}\cdot\boldsymbol{H}_0$ to γ in (84).

D. Source, with projection

$$\Delta = \sum_{i,j}\left(\nabla\cdot\boldsymbol{M}\right)_{i,j} - \sum_i\left(\gamma_i^{\mathrm{top}} + \gamma_i^{\mathrm{bot}}\right), \qquad S_{i,j} = -D\left[\left(\nabla\cdot\boldsymbol{M}\right)_{i,j} - \frac{\Delta}{n_X n_Y}\right]. \tag{85}$$

The divergence uses the isotropic nine-point stencil already employed for $\nabla\phi$. The projection is applied only in the all-Neumann case; it is dropped as soon as a Dirichlet face exists.

E. Collision and streaming

$$h^{\mathrm{eq}}_{k,i,j} = w_k\,\psi_{i,j}, \tag{86}$$

$$h^{*}_{k,i,j} = h_{k,i,j} - \left(\mathsf{M}^{-1}\mathsf{S}_h\mathsf{M}\right)_{k\ell}\left(h_{\ell,i,j} - h^{\mathrm{eq}}_{\ell,i,j}\right) + w_k S_{i,j}, \tag{87}$$

$$h_{k,i,j}(t{+}1) = \begin{cases} h^{*}_{k,i',\,j-e_{ky}}, & 0 \le j - e_{ky} \le n_Y - 1, \\ h^{*}_{\sigma(k),i',\,j_r} + B_{i'}\,w_k, & \text{otherwise}, \end{cases} \qquad i' = (i - e_{kx}) \bmod n_X, \tag{88}$$

with

$$j_r = \begin{cases} -(j - e_{ky}) - 1, & j - e_{ky} < 0, \\ 2n_Y - 1 - (j - e_{ky}), & j - e_{ky} \ge n_Y, \end{cases} \qquad B_i^{\mathrm{bot}} = 6D\gamma_i^{\mathrm{bot}}, \qquad B_i^{\mathrm{top}} = 6D\gamma_i^{\mathrm{top}}. \tag{89}$$

Setting $B \equiv 0$ degrades (88) to $\partial\psi/\partial n = 0$, which violates (49).

F. Field recovery

$$H_{x,i,j} = \bar{H}_x(t) + \frac{1}{\tau_h c_s^2}\sum_k e_{kx}h_{k,i,j}, \qquad H_{y,i,j} = \bar{H}_y(t) + \frac{1}{\tau_h c_s^2}\sum_k e_{ky}h_{k,i,j}. \tag{90}$$

The entire induced field, in both components, comes from the lattice. On a domain bounded in the field direction the leading terms vanish and (90) returns the total field.

G. Ordering within one time step

$\nabla\cdot\boldsymbol{M}$ and the projection Δ are non-local, so the field, magnetisation and source passes must each sweep the whole domain before the next begins. The h update itself remains inside the same time step and is executed once.

$$\underbrace{(90)}_{\boldsymbol{H}\text{ from }h} \;\to\; \underbrace{\text{MRSh}}_{\boldsymbol{M}\text{ from }\boldsymbol{H},\Omega} \;\to \underbrace{(84),(85)}_{\gamma,\,S} \to \underbrace{(82),(86)}_{\psi,\,h^{\text{eq}}} \to \underbrace{(87),(88)}_{\text{collide, stream}} \tag{91}$$

The magnetic forces and the f and g lattices are untouched and keep their present position in the loop.

H. Parameter selection

$$\varepsilon: \quad \text{minimise } \tau_{\text{relax}}(\varepsilon) = \max\left(\frac{1}{\varepsilon k_{\min}^2},\, \frac{\varepsilon}{c_s^2}+\frac{1}{2}\right) \implies \varepsilon \simeq \frac{c_s}{k_{\min}}, \tag{92}$$

$$\tau_B: \quad \tau_B \gg (\omega\tau_B)\,\tau_{\text{relax}} \qquad [\text{Eq. (54)}]. \tag{93}$$

A third, independent, physical constraint fixes the same scale. Eliminating the particle volume via $\tau_B = 3\eta V/k_BT$ and $\xi_0 = \mu_0 m H_0/k_BT$,

$$\frac{\varphi_p M_s}{H_0} = \frac{3\,\eta\,\varphi_p\,\xi_0}{\mu_0 H_0^2\,\tau_B}, \tag{94}$$

which must match the fluid being reproduced and is an *output* of the parameter choice, not an input. Since (93) and (94) both scale as τ_B^{-1}, they are satisfied together. Holding $\Omega\tau_B$ fixed under a change of τ_B requires the driving body force to scale as τ_B^{-1}.

I. Diagnostics

$$\Psi = \sum_{i,j}\sum_{k} h_{k,i,j} \qquad \text{bounded} \;\Leftrightarrow\; (49)\text{ holds; linear growth} \;\Rightarrow\; \text{no solution} \tag{95}$$

$$E_4 = \frac{|\Delta|/(n_X n_Y)}{|\nabla\cdot\boldsymbol{M}|} \qquad \text{fictitious charge injected by the projection} \tag{96}$$

$$\omega\,\tau_{\text{relax}} \qquad \text{validity of the single sub-step} \tag{97}$$

XIII. VERIFICATION PROTOCOL

With $\boldsymbol{M}$ frozen and the momentum solver disabled:

1. **Manufactured solution, all-Neumann.** Take $\psi = \sin(k_x x)\cos\big(k_y(y+\frac{1}{2})\big)$ with $k_y = m\pi/n_Y$ so $\gamma = 0$ at both walls, set $\nabla\cdot\boldsymbol{M} = \nabla^2\psi$, and confirm $E_1 \propto \Delta x^2$. The converged ψ must be *independent of* ε; this single test validates D, c_Φ, a and s_0 simultaneously, because only their combination (71) and the identity (34) can break ε-independence.
2. **Non-zero wall flux.** Repeat with $\gamma \neq 0$ to exercise (89). The exact solution of $\psi'' = \text{const}$ with prescribed end slopes is quadratic and must be reproduced to round-off.
3. **Dirichlet closure.** Impose ψ_w on two opposite faces; the same quadratic must be reproduced at second order, and E_4 must be identically zero because no projection is applied.
4. **Uniformly magnetised slab.** $\boldsymbol{M} = M_0\hat{\boldsymbol{y}}$ gives the exact result $H_y = -M_0$ (transverse demagnetising factor unity for a slab) with H_x undisturbed. A returned $\mathscr{C}M_0$ with $\mathscr{C} \neq 1$ localises the failure to (71).
5. **Mean-field consistency.** Verify $\langle\sum_k e_{kx}h_k\rangle_\Omega = 0$ to round-off. By the Corollary this must hold identically for periodic ψ; a non-zero value indicates the streaming closure has broken periodicity.
6. **Solvability.** Verify Ψ bounded and E_4 at round-off with γ from (44); contrast with $\gamma \equiv 0$, where Ψ must grow linearly at rate $\Delta t\sum_{\boldsymbol{x}} S$.